Module Code: CS1SE20

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Student Name	State your own contribution explicitly in parts or sections of the assignment report	Please confirm you have put 100% of your expected effort to your part; or indicate a percentage of the expected effort	Signature
Arkar Sithu Moe 32021041	Brief description of non-functional requirements, list of functional requirements, group discussions and formatting	100%	
Felix Possoz-Beech 32014331	Brief description of development approaches, list of non-functional requirements, group discussions and formatting	100%	
Asish Padhi 32013892	Description and evaluation of elicitation methods for finding requirements, group discussions and formatting	100%	
Chau Nguyen 32015995	Description and evaluation of elicitation methods for finding requirements	70%	
Frenci Hasanaj 32024674	Brief description of functional, reflection, conclusion, formatting, group discussions and formatting	85%	
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1. Introduction

There are problems with this case domain of the railway network services specifically the reservation system, but mostly the RRS Railway reservation system aims to make process of train transport easier such as scheduling, passenger information, reservation management, billing, and cancellations all these are key requirements that RRS is trying to implement into their system.

2. Analyse and compare various requirement elicitation approaches for gathering requirements for RRS

To gather the requirements for the software the RRS must use elicitation methods to gather data from stakeholders: the customers, the passengers, train operators, the government and Office of Rail and Road (ORR). There are multiple methods of elicitation; the 3 that will be covered will be brainstorming, interviews and use cases.

In brainstorming, a team is formed consisting of representatives from each stakeholder, where new ideas are generated and documented for a plan of action. This is quite useful to generate creative ideas due to the collaboration of the various backgrounds involved, bringing unique insight into problems. Due to ideas being discussed it also can allow for critique and ideas to add onto one another and is easier to merge ideas than other methods where numerous ideas may have difficulty merging due to participants being blind to other suggestions people have. The issue with this is that it can impact the ideas participants present through groupthink, a phenomenon where people avoid alternative ideas to the main idea discussed to conform; this can lead to less opinions being discussed or people being less open about critique due to fear of backlash as well as overlooking key errors due to less willingness to critique popular opinions out of judgement from the group. In the case of the RRS, a major flaw is that this method may struggle to consider the range of ideas the passengers will have as they are quite a large demographic and so it would be hard to involve them in a brainstorming team. This could be solved by taking a portion of the passengers to be involved in a team, however, a problem with this is that it can lead to biases especially since it may not account for various ideas people have.

In interviews, members of all stakeholders are interviewed and questioned about the issues and their needs. This is highly useful for planning as knowing the specific needs of all the stakeholders' aids in creating the most accurate requirements that are adaptable for all users. The primary issue that arrives from this method, however, is that there is bias between stakeholders, especially since there are many different groups of stakeholders. For instance, the opinions of the passengers and the government may differ greatly, making it so that the ideas generated are hard to merge; this can cause the requirements to be complex and difficult to work with. Also, this could lead to bias towards certain stakeholder's ideas due to some stakeholders having more authority than others. This issue also arrives between individuals within the same group, albeit on a smaller scale, as every individual has their own unique perspective, thoughts, and opinions about different subjects, making it so that interviewing them will give differing answers for the final solution. A good solution to this issue is narrowing down the people who are interviewed by selecting a particular few from each group; however, this introduces a new issue as individuals selected may not be representative of the majority in the group especially for groups that have

diverse backgrounds within them (. e.g., passengers). A solution to this issue can be through dividing the groups into subgroups (different backgrounds of passengers) and then randomly selecting participants for the interviews so that each subgroup is covered. Despite this solution, a downfall of this process is that it can be lengthy and would require more time to set up than other methods of elicitation for requirements.

In use cases, relevant files, and documents from past data to create requirements. Out of the three methods, this one is the best for producing results while interacting with the least amount of people. It runs the risk of neglecting individual stakeholder's needs. Primarily, the issue with using past information is that data and people change quickly, meaning that past data may not always be relevant to the present. For example, a surplus of passengers in the past can help create specifications, but the number of people causing an overflow can vary in this scenario. This is a major issue when considering the requirements relating to the number of reservations the system needs to hold. If all cases are used, then it can be difficult to quantify how many reservations are enough or how many reservations would be considered too many. With varying overflow cases, the true maximum would be difficult to gauge. To help gauge more specific details and make use cases more relevant to gather requirements for the RRS by using personas (personas are fictional users that are made to represent a group of people that hold similar views); this aids in better understanding customer needs by making fictional templates of people so that the target's needs are better met and can lead to more satisfactory results to the customer-centric approach. Multiple personas can be made to represent those from different demographics and habits and can help prioritize certain features over others in a use case to meet the personas' goals better and in turn meet the requirements for the customer. Although this is beneficial, this can have the disadvantage of the data used to form personas being biased, leading to inaccurate personas, which also runs the risk of stereotypical as not all users of a certain demographic will share the same views, and in some cases by focusing too much on the needs of individuals, personas can miss the broader picture.

3. Define and discuss the RRS' functional and non-functional requirements

Functional requirements describe in detail what a system should and (sometimes) should not do. It is crucial that these are written in a way that is as complete and consistent as possible so that misunderstandings do not occur. Functional requirements give users, stakeholders, and developers a mutual understanding about what to expect from the system.

Non-functional requirements specify how the system should run and do not directly contribute to the services of the system but add constraints to the services. These are usually of more significance than functional requirements because the system may not be approved by the public and government without non-functional requirements. (Sommerville, 2016)

The waterfall development approach has been chosen for this project. Waterfall requires a detailed up-front analysis and careful planning; the functional and non-functional requirements must be as detailed as possible to avoid errors. However, a superior alternative is AGILE, in AGILE a simple list of tasks to perform and detailed analysis is done just in time before each iteration to allow for early customer feedback. However, the size and complexity of the project meant waterfall was chosen. (Gaba, 2023)

Functional requirements for RRS application:

Account creation:

A user will be asked to login to their existing account or create a new account the first time they open the application. A user shall have to create an account to access the application (if they do not have one) by filling in their full name, gender, date of birth, phone number, email address, password, reconfirm password and agreeing to terms and conditions. A user can edit their profile such as username (set to full name by default), password (after confirming the authenticity of the user), adding billing information and rail cards.

Booking and tickets:

A user can view the timetables (peak, off-peak or both) of any station which shows live ticket prices (child and adult), permission to carry bikes and/or animals, disability details, departure and arrival times, cancellations, delays, strikes, seat types (window/table/individual/facing) and categories (first/standard/business class) available of every train from the respective station to any other station. A user can select filters such as cheap to expensive and other features included in the timetables. A user must choose the number of tickets (child and/or adult) when buying ticket(s) but disability details, seat types and categories are optional. A user must enter their billing information before every financial transaction if they have not already saved this on their profile.

Compensation and refund:

If a user has bought a ticket for a train which have been cancelled, the system will give the user two options: ask for a full refund or receive a free ride (same departure and arrival locations as the cancelled ticket) using any railway service in the next 24 hours. The system shall record delayed time (actual arrival time – expected arrival time) after the arrival of every train and give users a compensation fee depending on the price of the ticket and delayed time.

Staff permissions:

An authorized staff can request for detailed information on number of tickets sold, passengers on train, seat reservations and total sale for every train which is automatically calculated by the system. An authorized staff can edit timetables of respective train stations, add train schedules, and edit train status such as seat availabilities, price changes, train departure delays and cancellations in real time. An authorized staff can inform users of changes made to their tickets with regards to delays and cancellations.

(Trainline.com Limited, 2023)

Non-functional requirements for RRS application:

Performance:

A high throughput is necessary for when you have thousands of people trying to buy tickets, access timetables, etc., at the same time. Scalability ensures the app does not slow down at peak times one of these two methods must be used, if

horizontal is used servers can be switched on and off depending on demand, reducing running cost while vertical will have to be constantly on. The app is made to have low use so that it runs smoothly on less powerful devices, such as phones, and does not affect other apps on the user's device. Load testing is crucial to be done after any update when the least users are using the app to know how a system will handle large loads of users.

Reliability:

Fault tolerance. The way to ensure this is by using redundancy backup systems (inactive duplicates of critical parts). The application must be accessible anywhere and anytime meaning, for example, servers must be up 24/7 and the app must use low bandwidth. A high meantime between failures. A low meantime to recovery. No matter what happens the system will continue to function smoothly. Data integrity ensures data stays correct, consistent and unaltered. In the case of an RSR system this means, for example, ticket information is the same as when brought and across any device. Software should be reliable.

Security:

The app and the server encrypt everything to prevent sensitive data being stolen. One way of doing this is by using the web protocol HTTPS.

4. <u>Reflection on applications of SE approach to problem-solving</u>

During this coursework my colleagues and I had done a lot of research into many different aspects of user requirements and techniques of user interface such as how the process is done when making a user web application from the negotiating with the client and the stakeholders of what they want web application to be. Due to the 4-page limit, we have had to severely cut down on our amount, this means the quality is significantly less than what it could be. In the future to have an optimal report it would have to be 6 pages.

5. <u>Conclusion</u>

In conclusion, requirements are gathered for the RRS using the following elicitation methods: brainstorming, interviews and use cases. Functional and non-functional requirements are written as complete and consistent as possible to ensure stakeholders and developers alike understand them. For future improvements, use cases could be primarily used as it is more effective in gathering requirements generally.

References

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